

# Karit (Keith) Matanachai

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## EDUCATION

Princeton University, Princeton NJ (GPA: 3.97/4.00)

August 2022 - May 2026

### Bachelor of Science in Engineering (B.S.E.) in Computer Science

- **Honors:** Tau Beta Pi (Early Inductee) - **Top 12%** of Engineering Class, Independent Work Award in CS - **Top 8%**
- **Relevant Coursework:** Natural Language Processing, Data Structures and Algorithms, Computer Networks, Programming Systems, Computer System Design, Intro to Machine Learning, Probability and Stochastic Systems

## SKILLS

- **Programming:** Python, C++, Typescript, HTML/CSS, SQL, Node.js, React.js (Next.js), PyTorch, Numpy, Java, Swift, C
- **Tools:** AWS, Docker, PostgreSQL, Jupyter Notebooks, GCP, Git, Tailwind CSS, shadcn, Drizzle ORM, Supabase, Sentry

## RELEVANT EXPERIENCE

### Software Engineering Intern - Amazon Web Services & Amazon AGI Labs (Boston, MA)

June 2025 - August 2025

- Re-architected a data processing pipeline for Amazon Speech Products (Alexa and Nova Sonic) in C++ by adding an abstraction layer that enabled modular and reusable components, reduced code duplication, and improved testability **without increasing latency (P90, P95, and P99)**; adopted by 3 internal teams
- Proactively diagnosed and fixed a systemic speech pipeline defect (around **15%** of test and production failures); delivered a base-package patch, rollout docs, and drove adoption across 5+ teams / 15+ engineers

### Software Engineering Intern - CommandBar (YC S20) (San Francisco, CA)

June 2024 - August 2024

- Optimized RAG pipeline by performing embedding model A/B testing, deploying via AWS Sagemaker along with Elasticsearch modifications; **reduced AI agent fallbacks by 30%** through improving relevant document retrieval
- Engineered an end-to-end AI agent evaluation Python pipeline, including real-time score monitoring and alerts for proactive regression testing and detection of performance issues, **leading to a 15% increase** in agent quality
- Implemented LLM observability for the AI agent product in Python, enhancing tracking for LLM workflows, which resulted in a significant reduction in debugging time and improved overall AI agent reliability
- Created cost-tracking analytics for custom AI agent workflows using Clickhouse, Metabase, and SQL queries, enabling precise quoting for prospective customers and cost reductions by identifying prompt and model optimizations

## TECHNICAL PROJECTS

### Full-Stack Developer - Curbon

July 2024 - Present

- Engineered an MVP platform to help consumers and corporations understand the environmental impacts of clothing purchases; built with Next.js, Supabase Auth for authentication, Drizzle ORM, and PostgreSQL for the database
- Developed key features allowing businesses to input manufacturing details, receive environmental impact reports, and display this information to consumers; set to launch with the University of California, San Diego in Summer 2025

### Predicting Prime Editing Efficiency Using ML - Fall Research Advised by Yuri Pritykin

Sep 2024 - Jan 2025

- One of 8 recipients (out of 98) of Princeton's Exemplary Independent Work Award in Computer Science
- Trained ML models using CNN, GRU, Attention & LSTM architectures to predict CRISPR-based prime editing efficiency
- Performed interpretability analysis with DeepSHAP on deep learning models to determine feature importances

### Third Place, 2024 COSCON – Princeton's Computer Science Contest

November 2024

- Third place among teams competing in COSCON, an annual competition hosted by Princeton's ACM student chapter

## LEADERSHIP

### Academic Life and Learning Consultant - The McGraw Center for Teaching & Learning

December 2023 - Present

- Provide academic coaching to students through strategic planning tailored to their courses and extracurriculars
- Facilitate course-specific workshops on study strategies and exam preparation, engaging over 30 students per session

### Social Chair - Princeton Computer Science Council

August 2024 - Present

- Work closely with department faculty and staff to improve the experience for undergraduates by hosting department-wide social events, including friendly competitions with prizes, trivia nights, and study breaks